

ECN 712 Fall 2025 EE Schlee
Problem Set XI, Due Wednesday, 4 December, at midnight.

I sprinkle Question 2 with some parenthetical questions. Don't turn in your answers to them, but I urge you to convince yourself you understand the answers on your own.

1. Analyze the game from problem set VIII, #3 (first-price sealed-bid auction with two bidders and complete information) from the viewpoint of the Debreu-Fan-Glicksberg theorem on existence of Nash equilibrium. Explain which assumptions the game (or some *innocuous modification of it*) satisfies and which it does not. (By “innocuous” I mean things such as relabeling the strategies or deleting strictly dominated strategies in the original game.)
2. For $i = 1, 2$, player i has an action set A_i , a nonempty compact real interval, and a continuous utility representation $U_i : A_1 \times A_2 \rightarrow \mathbb{R}$. Let $G = (\{1, 2\}, \{A_1, A_2\}, \{U_1, U_2\})$ be a strategic form game, and G_{St} the “Stackelberg” game derived from G in which player 1 first chooses $a_1 \in A_1$; after observing a_1 , player 2 chooses $a_2 \in A_2$. Player i 's preferences over *outcomes* in G_{St} are represented by U_i .

- (a) Let R_i be player i 's best reply correspondence in G . Explain why it is nonempty for every $a_j \in A_j$, $j \neq i$. Identify a sufficient condition on U_i for R_i to be single-valued.

In what follows assume directly that each R_i is single-valued—implying that R_i is continuous (by Berge's Maximum Theorem)—and that $NE(G)$ and $SPNE(G_{St})$ are nonempty.

- (b) Compare player 1's welfare in $NE(G)$ and $SPNE(G_{St})$. (Note that neither set need be a singleton, so it might not be possible to say one way or the other.) Is there a difficulty extending your conclusion to the case in which $R_2(\cdot)$ is not single-valued? Explain.
- (c) (Bulow-Geanakoplos-Klemperer) Suppose that G is a Cournot game whose inverse demand $\tilde{p}$ satisfies the textbook monopoly assumptions; and which has compact strategy sets (say, because the inverse demand is positive if and only if aggregate output is less than some $\bar{Q} > 0$); and with each firm i having a CRS technology with constant unit cost of k_i , where $\tilde{p}(0) > k_1 > \tilde{p}(q_2^m) > k_2 > 0$, where q_2^m is firm 2's (unique?) monopoly output. Let

$$S(q_1) = \int_0^{R_2(q_1) + q_1} p(y) dy - k_1 q_1 - k_2 R_2(q_1), \quad (1)$$

the welfare in G indexed by firm 1's output (why is this welfare?). Suppose in addition that R_2 is differentiable. (i) Identify a condition on R_2 which ensures that $S'(0) < 0$ if k_1 is close to $\tilde{p}(q_2^m)$. Since $NE(G) = \{(0, q_2^m)\}$ if $k_1 \geq \tilde{p}(q_2^m)$; and $q \gg 0$ in any NE of G when $k_1 < \tilde{p}(q_2^m)$ (why for each?); the conclusion implies that equilibrium social welfare can decrease when firm 1's technology improves. (ii) Explain economically.

3. Consider the game described in MWG, Exercise 9.B.9. Identify every subgame-perfect Nash equilibrium (SPNE) *outcome* of the twice-played stage game. For each SPNE *outcome* you identify, write down a SPNE *strategy* that supports it. How do the set of SPNE outcomes of the repeated games compare with the set of NE of the once-played game?
4. *Evaluate*: “In a finitely repeated prisoners' dilemma, the strategy profile in which each player's strategy is ‘defect in every period, no matter what the history’ is a dominant strategy equilibrium.” Prove it if it is true, give a counterexample if it is false. I suggest you start with the 2-date case. (The definition of a dominant strategy equilibrium is given in my notes on Strategic Form Games.)
5. MWG, Exercise 12.D.1, parts (a) and (b). In addition, for part (a), when is it easier to sustain collusion, when $\gamma < 1$ or $\gamma > 1$? Explain economically.¹

¹For part (b), it is easy to show that the game ends in finite time with probability 1 (equivalently, the probability that the game never ends is 0). This example shows that a *known* infinite horizon is not necessary to get collusion in a repeated Bertrand game.